

LESSON 1.10

Rational Functions and Holes

AP Precalculus · Unit 1: Polynomial and Rational Functions

THINK ABOUT IT

Suppose your GPS loses signal for exactly one moment — at $t = 5$ seconds — but your position function is defined everywhere else.

What we know

- ▶ The output at $t = 5$ is missing (undefined).
- ▶ Outputs for t near 5 are all well-defined.
- ▶ Those nearby outputs cluster around one value L .

What this means

- ▶ We can predict the “missing” value: L .
- ▶ This is exactly a **hole** in the graph.
- ▶ We write: $\lim_{t \rightarrow 5} r(t) = L$

Question: How is this different from a vertical asymptote?

DECISION CHART

Let $r(x)$ have a shared zero at $x = a$. Let m = numerator multiplicity, n = denominator multiplicity.

HOLE at $x = a$ (if $m \geq n$)

- ▶ Common factor cancels.
- ▶ Graph has a missing point.
- ▶ $\lim_{x \rightarrow a} r(x) = L$ exists and is finite.
- ▶ Example: $\frac{(x-2)(x+3)}{x-2}$
hole at $x = 2$

VA at $x = a$ (if $m < n$)

- ▶ Denominator “dominates” near $x = a$.
- ▶ Outputs grow without bound.
- ▶ $\lim_{x \rightarrow a} r(x) = \pm\infty$.
- ▶ Example: $\frac{x-2}{(x-2)^2}$
VA at $x = 2$

4-STEP PROCEDURE

Steps

1. **Factor** numerator and denominator.
2. **Identify** shared zeros and multiplicities.
3. **Cancel** the common factor.
4. **Substitute** $x = c$ into simplified form to get L .

Example 1: $r(x) = \frac{x^2 - 4}{x - 2}$

1. Factor: $\frac{(x - 2)(x + 2)}{x - 2}$
2. Shared zero $x = 2$; $m = n = 1$, so **hole**.
3. Cancel $(x - 2)$: simplified $= x + 2$
4. $L = 2 + 2 = 4$

Hole at $(2, 4)$ and $\lim_{x \rightarrow 2} r(x) = 4$

WORK WITH A PARTNER

Analyze $f_2(x) = \frac{x^2 - 1}{(x - 1)(x + 4)}$.

Your Tasks

1. Factor the numerator and write the full factored form.
2. Identify any shared zeros. For each, compare m and n and classify as hole or VA.
3. Find the coordinates of any holes (cancel, then substitute).
4. Identify any additional vertical asymptotes.
5. Write the limit statement for any holes.

Be ready to share: coordinates of the hole, the VA equation, and the limit notation.